Component	Cost	Link	Units	Size
Buck IC	$0.89	Buck IC Data Sheet	1	
Inductor	$1.10	Inductor Data Sheet	1	.224’”x.205’W
Capacitor	$1.35	10 MicroF Data Sheet	1	0603
	$0.36	22 MicroF Data Sheet	1	
	$0.14	100 nF Data Sheet	1	
Total Cost	$3.84			

Some incomplete/abandoned notes for choosing components after a while I stopped adding to it when I got into Kicad. Most of the accurate stuff about the components is on Kicad.

Buck Converter Components

Buck IC
Adjusts duty cycle of switches and works with feedback loop
Input Voltage ~10-15V *
Output Voltage 5V
Output Current I out =~ 2A
Switching Frequency Fsw 500kHz**
Ripple Voltage 1-5%

Selection Data:

3.8V(min)- 32V(max) input voltage
5V fixed output
1.1Mhz switching frequency
20mV Output Voltage Ripple

Inductor
*Saturation Current- Make sure they don't oversaturate at expected output current
2.2-10 Micro Henrys
DC Current Rating 2.7+
Inductor Current Ripple between .6-.9
Peak Current 2.3-2.45
Saturation Current (2.76-3.3075)+

Selection Data:
Inductor Current Ripple .86
Peak Current 2.43A
Inductance L = 5.6 Micro Henries
Rated Current 6.4A
Saturation current 12.8 A
Output Capacitor
DC Biasing- Capacitor does not stay at specific capacitance designed to be at.
Capacitance drops by some percent when voltage applied across it.
Match Output Capacitator type to noise frequency

Input Capacitor
Low ESR
RMS current rating must be higher than RMS input current or greater than half maximum load current
Tantalum Capacitor must be surge protected
Ceramic Capacitor > 10microFarads Sufficient

38.4+ Voltage Rating

Selection
10microF
Low ESL
50V Rated

LV Standards
Power drives Signals which flow left to right

Connect signals with wires
Label wires with net labels
Connect power with power signals
Global labels in case where you must "teleport" wires

0603 Sizing for Resistors and Capacitors
Hand soldering version for footprints
Capacitors rated for high enough voltage (50% Margin)
Multilayer ceramic capacitors

Pin requirements
Pin placements must be outside symbol(Not connected to edge)
Pins on 100mil grid with 100-300mil lengths, 50mil increments
Group Pins by Function

Reference	Reference designator [U?]
Value	Part Number for specific components Value for generics (Resistors and capacitors)
Footprint	Footprint, if component has specified footprint
Datasheet	Link to datasheet for component
Description	Description of the component
Digi PN	Digikey Part Number
Digi URL	Digikey URL
MFG PN	Manufacturer Part Number
MFG	Manufacturer Name

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